YUHANG JIANG

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Summary

- Double-degree MSc in Computer Science (University of Trento) and Data Science (SUPSI), EIT Manufacturing programme. Research intern at Huawei Pisa, working on multimodal agents and LLM post-training.
- Two first-author journal papers accepted in 2026: IEEE Transactions on Geoscience and Remote Sensing and TMLR.
- GeoSelect sets the training-free state of the art on RRSIS-D and RISBench. DeliverMem sets the state of the art on DMV-Bench and leads published memory-augmented agents on MemLens.
- Research focus: multimodal agents, LLM post-training, embodied AI and visuomotor policies. Several projects were run independently, from problem definition to paper.

Education

EIT Manufacturing Master & Doctoral School (Double Degree)
Data Science and AI for a Competitive Manufacturing · Scholarship: full tuition waiver + 10,000 EUR/yr

- **Università di Trento (UNITN), Italy** Sep 2023 - Mar 2026
  MSc, Computer Science · GPA 98/110 · Degree awarded Mar 2026
- **University of Applied Sciences and Arts of Southern Switzerland (SUPSI)** Sep 2023 - Dec 2026
  Master of Engineering (MSE), Data Science · GPA 5.1/6.0 · Thesis defended Mar 2026, diploma expected Dec 2026

Anhui University (Project 211), China Sep 2019 - Jun 2023
BEng, Intelligent Science and Technology · Average 84.34/100
Scholarships: Academic Science and Technology Scholarship, First Prize (2021); Outstanding Academic Performance, Second Prize (2022)

Research Experience

Huawei Pisa Research Center Apr 2026 - Oct 2026
Research Intern Pisa, Italy
Independent research: memory for multimodal agents (first-author paper under review)

- **Located the bottleneck at the delivery step with controlled experiments.** With the retrieved set fixed and one variable changed at a time, delivering the original images adds +13.87 points on Qwen3-VL-8B, while perfect retrieval adds only +2.31. The delivery gain grows to +17.92 on Qwen3-VL-32B.
- **Proposed DeliverMem, a training-free delivery method that leaves the memory store unchanged.** It keeps the original modality, readable identifiers and temporal position, with a small retrieval-side adaptation. It is the **state of the art on DMV-Bench**, ahead of the best prior method at every setting by 7.63 to 22.62 points. On MemLens it leads the strongest published memory-augmented agent by at least 8.20 points at all four context lengths. At MemLens 32K with dev-fitted selection it **surpasses GPT-5.4 and matches Gemini-3.1-Pro**, while delivering about 2,400 tokens, a tenth of what they read.

Business project: LLM post-training for in-cabin AI (sole owner)

- **Data**: checked where each answer field could be copied from in the inputs, then redesigned the training format. Added an entity-rendering step and removed a segment that repeated the answer. Replaced augmentation that broke copied spans with whole-record rewriting, verified by fingerprints.
- **SFT**: LoRA fine-tuning of Qwen3-4B / 8B and Qwen3.6-35B-A3B (MoE) with ms-swift. Implemented a set-prediction loss (minimum-permutation CE), hard-case weighted SFT and rejection-sampling fine-tuning.
- **Preference optimization and RL**: DPO and triplet-level DPO (preference pairs split by Hungarian alignment), GRPO and DAPO. Designed a hierarchical GRPO reward with format, schema, triplet F1, direction, relation-class and process terms.
- **Evaluation**: built a triplet evaluator (Hungarian matching, placeholder equivalence, relation-synonym normalization) with batched vLLM inference. Audited and repaired the business memory evaluation set, replacing LLM-based pairing with a deterministic memory-level metric.

Fondazione Bruno Kessler (FBK) Mar 2025 - Mar 2026
Research Intern Trentino-Alto Adige, Italy

- **Proposed Active Instance Verification and built its first benchmark, PInVerify**: 3,000 episodes over 18 categories in photorealistic 3D scenes, with trap views and unreachable sectors.
- **Evaluation protocol**: positive, same-category and cross-category pairs balanced 1:1:1 to isolate calibration bias, with steps-to-decision and navigation-failure metrics and disjoint train and test pools. Confidence intervals placed the bottleneck in detection.
- **Two baseline tracks**: a training-free modular pipeline, and an end-to-end Qwen3-VL-4B + LoRA agent post-trained on 22K self-built CoT samples with SFT, DPO, GRPO and GSPO, rising from 70.6% to **85.6%**.

Chengdu Jiaoda Guangmang Technology Co., Ltd. Jul 2022 - Sep 2022
Algorithm Intern Chengdu, China

- Industrial anomaly detection: GAN-based image reconstruction and object detection for defect inspection on high-speed rail components.

Publications

"GeoSelect: Spatial-Program Execution for Training-Free Referring Remote Sensing Image Segmentation"

IEEE Transactions on Geoscience and Remote Sensing (TGRS), 2026 · First author · IEEE Xplore · Project Page · arXiv:2607.03869
Recasts referring segmentation as executing typed spatial programs, with no referring supervision. Reaches 58.86 / 55.27 mIoU on RRSIS-D / RISBench, over 2× / about 1.7× the best prior training-free results.
Yuhang Jiang, Guohui Deng*, Miaozhong Xu, Chao Ruan, Jinling Zhao, Linsheng Huang
"The Scissors Effect: When Resize-Based Input Diversity Helps or Hurts Transfer Attacks"
Transactions on Machine Learning Research (TMLR), 2026 · First author · Project Page · arXiv:2606.22516 · OpenReview
Resize-based input diversity is a common default in transfer attacks, but it hurts transfer from adversarially trained surrogates. Using it without checking overstates the robustness of the target model.
Yuhang Jiang, Xiaojing Chen
"PInVerify: An Offline Embodied Benchmark for Active Instance Verification"
Poster, **FMEA Workshop @ CVPR 2026** · Sole author · Project Page · arXiv:2605.30639 · OpenReview
Introduces the Active Instance Verification task and its first benchmark. Active viewpoint selection gives no reliable gain over random views, and detection quality sets the ceiling.
Yuhang Jiang · Master's thesis project
"When Dynamic Scaling Fails: A Scale–State Feedback Governs Recurrent-State Quantization in Linear-Attention and SSM LLMs"
LCFM Workshop @ NeurIPS 2026 (non-archival) · Sole author · Project Page · arXiv:2610.15139 · OpenReview · Code
INT4 that is lossless at the start destroys recall a thousand steps later: the quantizer sits inside the recurrence, so containing the scale beats locating outliers.
Yuhang Jiang

Preprints and Other Publications

"GeoScope: Training-Free Program Execution beyond the Counting Horizon for Ultra-High-Resolution Remote Sensing Question Answering"
arXiv:2611.12699 · Co-first author · Project page · under review
Frozen vision–language models undercount aerial objects past a measured horizon of about five per view; splitting views by that limit adds 16.3 points on XLRS-Bench regional counting, untrained.
Guohui Deng*, **Yuhang Jiang**, Chengjun Wang, Zhongwen Hu, Zhaocong Wu, Linsheng Huang, Jinling Zhao, Chao Ruan, Miaozhong Xu, Chi Chen
"Retrieved but Not Delivered: Multimodal Memory Delivery for Long-Term Agents"
arXiv:2609.32590 · Project page · under review
The bottleneck of multimodal memory is delivery, not retrieval; training-free DeliverMem leads the strongest published methods on MemLens and DMV-Bench.
Yuhang Jiang, Qingwei Liao, Kaize Yin, Xingling Liu, Luca Cuomo, Silvio Bacci
"Median Temporal Ensembling: Training-Free Robust Aggregation for Action-Chunked Visuomotor Policies"
arXiv:2609.27167 · Sole author · Project page · under review
A per-coordinate median in place of the weighted mean (one line, no retraining) is never worse in 25 settings for ACT and Diffusion Policy and significantly better in 15.
Yuhang Jiang
"A Circuit, Not The Circuit: Non-Unique Causal Localisation of the Mamba-2 State Sink"
arXiv:2606.00930
Names the state sink, the state-space counterpart of the attention sink. A nearly disjoint set of heads matches the causal effect of the probe-selected ones, so a probe finds a circuit, not the circuit.
Yuhang Jiang, Bowen Zhang
"Refit the Probe: Single-Direction Ablation Is Not a Necessity Test"
arXiv:2606.00926
A probe refitted on the ablated activations recovers its accuracy, so single-direction ablation cannot establish necessity. Subspace erasure, checked against random subspaces of the same size, separates the layer where a variable is easiest to read from the layer where erasing it hurts most.
Yuhang Jiang

Awards and Competitions

- **Kaggle Competitions Expert** (all solo): Silver, March Machine Learning Mania 2025 (64 / 1,727); Bronze, BirdCLEF+ 2025 (199 / 2,025); Bronze, BirdCLEF 2024 (70 / 974).
- **Mathematical Contest in Modeling (MCM), Finalist** (top 1%), 2021. Team leader; responsible for modelling and implementation.

Skills

- **Research areas:** multimodal LLMs, agent memory, LLM agents, RAG, embodied AI, mechanistic interpretability.
- **LLM post-training:** SFT, DPO, GRPO / GSPO / DAPO, RFT, RL reward design, LoRA / PEFT.
- **Frameworks and programming:** PyTorch, Transformers, vLLM, ms-swift; scikit-learn, MPI / OpenMP (HPC); Python, C/C++, Matlab, R.
- **Service and languages:** Reviewer, ICLR 2027, ICRA 2027, FMEA @ CVPR 2026. English (fluent, TOEFL 102); Chinese (native).